Exercise Sheet 1

Big Data in WS 2026/27

Date of issue: 21.10.2026 Deadline: 22.10.2026

The exercises marked with ★ are relevant for admission to the examination and must be submitted in electronic form in Moodle by the submission deadline.

Exercise 1.1 (*Sum of kth powers*).

Repeated multiplication of a number by itself corresponds to raising that number (the base) to an exponent, say k. The addition of such kth powers, each with a different base, is a common process in various areas of mathematics. A prominent example where such sums of powers of numbers occur is statistics where certain aspect of data are formulated in terms of powers of the available data points. In this exercise, we consider the following problem. Given an integer k and a sequence of n numbers

$$X = [x_1, x_2, \ldots, x_n],$$

compute the sum of the kth powers of X given by

$$y_k(X) := \sum_{i=1}^{n} x_i^k.$$

Formulate the computation of $y_k(X)$ by using the functions map and reduce.

Exercise 1.2 (*kth frequency moment*).

Sequences of elements are sometimes characterized by a single value that quantifies an interesting property of the sequence. The following definition characterizes a situation arising from data streaming. A sequence

$$S = [a_1, a_2, \ldots, a_n]$$

of n elements $a_i \in M := \{1, 2, \ldots, m\}$ is called a *data stream*. Given an integer $k \geq 0$ and a data stream S, the *kth frequency moment of* S is defined by

$$F_k(S) := \sum_{i=1}^{m} f_i^k,$$

where f_i is the number of times the element $i \in M$ occurs in the stream S. Similarly, define another frequency moment by

$$F_\infty(S) := \max_{1 \leq i \leq m} f_i.$$

Let

$$S = [1, 1, 2, 1, 2, 3, 1, 2] \tag{1}$$

denote the (finite) data stream with $n = 8$ elements taken from the set $M = \{1, 2, 3, 4\}$, which corresponds to $m = 4$.

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a) Compute the frequency moments $F_k(S)$ of that data stream for $k = 0, 1, 2$ and $k = \infty$. Interpret the meaning of these four moments by stating what different properties of a data stream are represented by these four moments.

b) Given an integer $k \geq 0$ and a data stream S, formulate the computation of $F_k(S)$ by using the functions map and reduce.

Exercise 1.3 (*Polynomial evaluation*).

In addition to the higher-order function

$$\mathsf{map}\left(\begin{bmatrix} \square \\ \square \\ \square \\ \blacksquare \end{bmatrix}, f \right) = \begin{bmatrix} f(\square) \\ f(\square) \\ f(\square) \\ f(\blacksquare) \end{bmatrix},$$

where the second input represents some function

$$f : \mathcal{X} \to \mathcal{Y},$$

some programming languages offer another version of map that takes two lists, L_1 and L_2, as input. This version, $\mathsf{map}(L_1, L_2, f)$, takes some function

$$f : \mathcal{X} \times \mathcal{X} \to \mathcal{Y}$$

as a third input and applies f to individual pairs of elements taken from the same position of each of the two lists. That is, a visual representation of that map version is given by

$$\mathsf{map}\left(\begin{bmatrix} \square \\ \square \\ \square \\ \blacksquare \end{bmatrix}, \begin{bmatrix} \square \\ \square \\ \square \\ \blacksquare \end{bmatrix}, f \right) = \begin{bmatrix} f(\square,\square) \\ f(\square,\square) \\ f(\square,\square) \\ f(\blacksquare,\blacksquare) \end{bmatrix}.$$

Use a combination of the two versions of map and the function reduce to compute the value of a polynomial

$$p(x) = p_0 + p_1 x + p_2 x^2 + \cdots + p_n x^n$$

of degree n evaluated at a point x.

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